

VIVEK RAJ SINGH

Bhopal, Madhya Pradesh

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Summary

AI and Systems Engineer building production-grade intelligence and distributed systems. Hands-on experience developing ML orchestration frameworks in Rust, optimizing algorithms for real-world impact to improve agricultural yields, and deploying complex pipeline APIs. Open-source contributor driving scalable architecture, reliability, and modern orchestration paradigms.

EDUCATION

VELLORE INSTITUTE OF TECHNOLOGY

2023 – 2027

B.Tech - Computer Science and Engineering (Cloud Computing) - **CGPA - 8.52**

Bhopal, Madhya Pradesh

Sunbeam English School, Bhagwanpur

2020 – 2022

Higher Secondary: **75.2%** (2021–2022) | Matriculation: **93.2%** (2019–2020)

Varanasi, U.P.

TECHNICAL SKILLS

- **Languages:** Python, Rust, C++, HTML, CSS
- **Databases/Tools:** SQL, MongoDB, Git/GitHub, Docker, FastAPI, Flask
- **Deep Learning/ML:** PyTorch, TensorFlow/Keras, Scikit-learn, YOLOv5/YOLOv8, ResNet50
- **Libraries:** OpenCV, Pandas, NumPy, Matplotlib
- **Core Concepts:** Distributed Systems, Systems Orchestration, Quantitative Risk (VaR), API Design

PROJECTS

Open Source Contributor – MoFA Framework | Rust, Distributed Systems

Jan 2026 – Present

- Designed and implemented a **Unified Inference Orchestrator** with a **runtime model-adaptor registry** for seamless local/cloud routing and dynamic capability negotiation.
- Implemented a resilient **circuit breaker & fault-tolerance layer** to ensure reliability across heterogeneous agent nodes.
- Engineered a **memory-aware admission controller** for optimized inference scheduling and resource limits.
- Actively driving core architecture decisions, opening and merging **5+ production PRs** across core repositories.

Stochastic Storage VaR Engine 🐙 | PyTorch, statsmodels, NumPy, Monte Carlo, FastAPI

- Architected an ML framework forecasting hardware degradation as a stochastic process using 730 days of synthesized, statistically rigorous telemetry data.
- Engineered an **Ensemble Forecasting pipeline** combining custom PyTorch LSTMs (Cosine LR, Huber Loss) and ARIMA models for extreme baseline efficiency.
- Built a quantitative risk engine performing **10k Monte Carlo simulations** to calculate severe failure bounds (VaR_{95} , $CVaR_{95}$); implemented **Isolation Forests** to detect security anomalies.
- Exposed models via highly-efficient thread-safe Python bundles capable for seamless ingestion into external REST APIs.

AgroVit – Smart Agriculture Platform 🐙 | TensorFlow/Keras, Flask, scikit-learn, ML

- Built an ML-powered agriculture platform with **3 modules: 10-class CNN** for pest detection, **ensemble model** (SVM, Random Forest) for crop recommendation, and NPK-based fertilizer optimization.
- Trained a CNN using **TensorFlow/Keras** with image augmentation; implemented a **VotingClassifier** combining 13 models for crop prediction based on soil/environmental features.
- Deployed a production **Flask REST API** with thread-safe model loading, image preprocessing pipelines, and a responsive frontend for real-time inference capabilities.

EXPERIENCE

BondsPe Financial Services Private Limited

Nov 2025 – Feb 2026

Data Analyst Intern

Remote

- Analyzed bond market data and performed yield comparisons (FD vs Bonds) for fixed income investment research.
- Applied credit risk models including probability of default (PD) estimation and credit spread analysis for issuer risk assessment.

CERTIFICATIONS

- Completed the **Google IT Support Professional Certificate** covering system administration, networking, and security fundamentals.
- Successfully completed **Infosys Springboard AI Internship**, covering end-to-end AI and ML concepts.
- English Proficiency Certification: **C1 Advanced** (EF SET).